



**COURSE:PROBABILITY & STATISTICS
MT-331**

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CHI-SQUARE DISTRIBUTION

The chi square test denoted by χ^2 . The Chi-square test is a useful measure of comparing experimentally obtained result with those expected theoretically & based on the hypothesis it is used as a test statistics in testing a hypothesis that provide a set of theoretical frequencies with which observed frequencies are compared.

The χ^2 -test was first used in testing a hypothesis that provide a set of theoretical frequencies with which observed frequencies are compared.

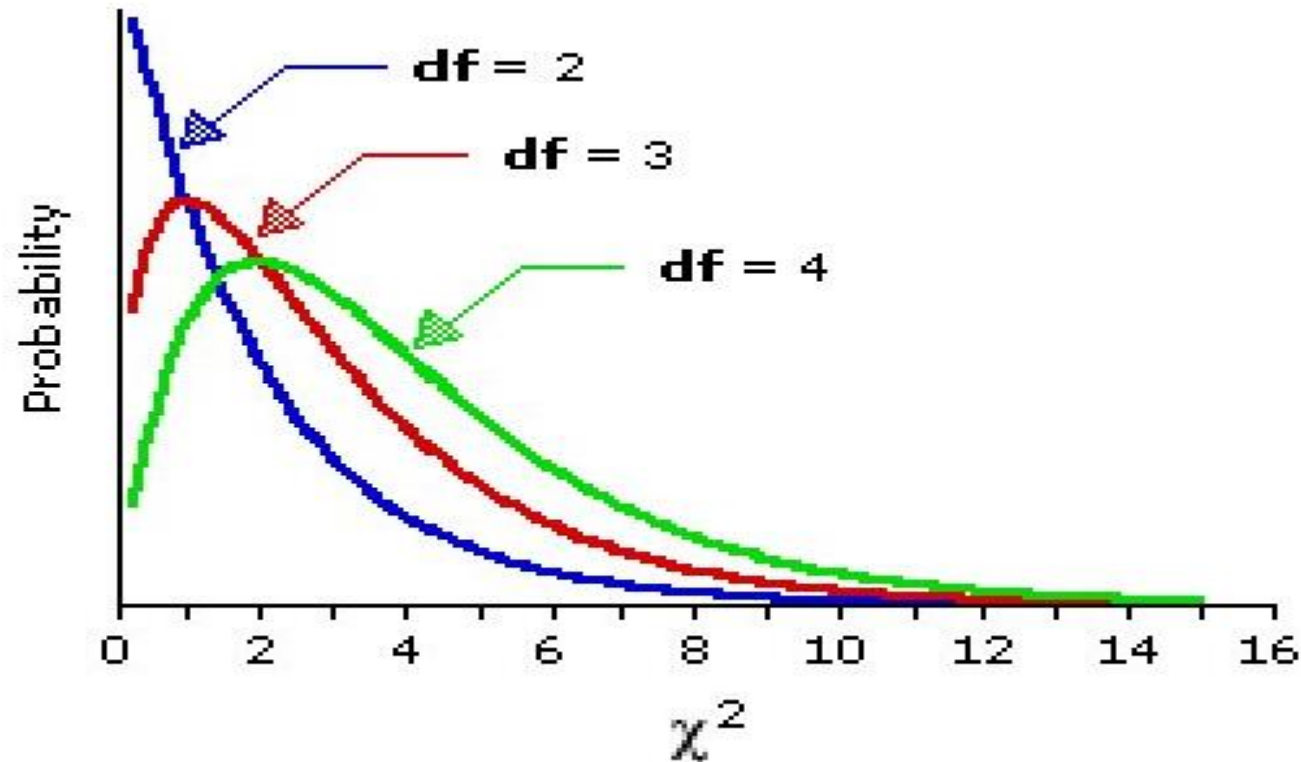
The χ^2 test was first used in testing statistical hypothesis by Karl pearson. it is defined as

$$\chi^2 = \sum_{i=1}^n \left(\frac{(o_i - e_i)^2}{e_i} \right)$$

Where o_i is the frequency of the i^{th} event.

e_i is the expected frequency of the i^{th} event

Chi square maybe skewed to the right or with a long tail toward the large values of the distribution. The overall shape of the distribution will depend on the number of degrees of freedom in a given problem.

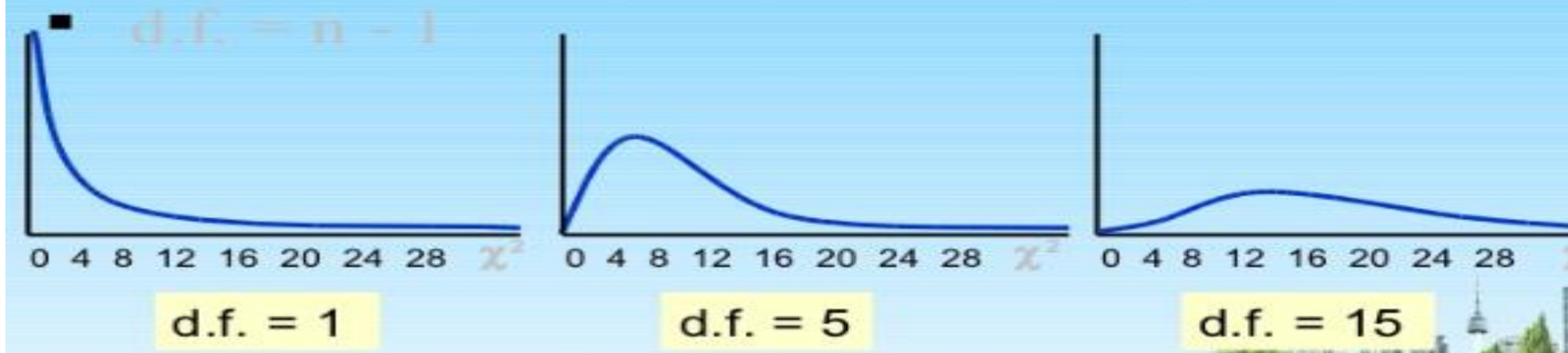


Properties:

1. The χ^2 distribution is continuous.
2. The χ^2 distribution is unimodal.
3. The χ^2 distribution is always positive (>0).
4. The χ^2 distribution mean = ν .
5. The χ^2 distribution variance = 2ν .
6. For small ν ($\nu < 10$), the distribution is highly skewed to the right (positive).
7. As ν increases the χ^2 distribution becomes more symmetrical about ν (the mean).
8. As the degrees of freedom increase, the chi square curve approaches a normal distribution.
9. As the degrees of freedom increase, the symmetry of the graph also increases.
10. Finally, It may be skewed to the right, and since the random variable on which it is based is squared, it has no negative values. As the degrees of freedom increases, the probability density function begins to appear symmetrical in shape.

The Chi-square Distribution

- The chi-square distribution is a family of distributions, depending on degrees of freedom:



Uses Of Chi-square Test

- 1-Test of Independence
- 2- Test of homogeneity or a test for a specified Standard deviation

Test for Independence (Categorical Data)

The [Chi-Square test of independence](#) is used to determine if there is a significant relationship between two nominal (categorical) variables. The frequency of each category for one nominal variable is compared across the categories of the second nominal variable. The data can be displayed in a contingency table where each row represents a category for one variable and each column represents a category for the other variable.

Null hypothesis: Assumes that there is no association between the two variables.

Alternative hypothesis: Assumes that there is an association between the two variables.

Hypothesis testing: where a test statistic is computed and compared to a critical value. The critical value for the chi-square statistic is determined by the level of significance (typically .05) and the degrees of freedom. The degrees of freedom for the chi-square are calculated using the following formula: $df = (r-1)(c-1)$ where r is the number of rows and c is the number of columns. If the observed chi-square test statistic is greater than the critical value, the null hypothesis can be rejected.

The Chi-square test procedure can also be used to **test the hypothesis of independence** of two variables.

- The observed frequencies are presented row wise as well as column wise and is classified as contingency table. A contingency table with r -rows and c -columns is referred to as an $r \times c$ table.
- To test the null hypothesis, H_0 of independence, the general formula for computing the expected frequency of any cell is defined as;

$$\text{Expected Frequency} = \frac{(\text{Row Total}) * (\text{Column Total})}{\text{Grand total}}$$

- To test the null hypothesis use chi-square test with $\nu = (r - 1) (c - 1)$ **degree of freedom**.

$$\chi^2 = \sum \left(\frac{(o_i - e_i)^2}{e_i} \right)$$

SIX STEPS FOR TESTING A HYPOTHESIS

- STEP 1 State null hypothesis
- STEP 2 State alternative hypothesis
- STEP 3 Select a level of Significance
- STEP 4 Select the test statistic
- STEP 5 Formulate the decision rule
- STEP 6 Take a decision

10.90 In an experiment to study the dependence of hypertension on smoking habits, the following data were taken on 180 individuals:

	Non-Smokers	Moderate Smokers	Heavy Smokers
Hypertension	21	36	30
No Hypertension	48	26	19

Test the hypothesis that the presence or absence of hypertension is independent of smoking habits. Use a 0.05 level of significance.

Solution:

H_0 : The Hypertension & smoking habit is independent of each other.

H_1 : The Hypertension & smoking habit is dependent of each other.

Level of Significance: $\alpha = 0.05$

Test Statistics: $\chi^2 = \sum \left(\frac{(o_i - e_i)^2}{e_i} \right)$ $e_i = \frac{(\text{row total}) * (\text{column total})}{\text{Grand total}}$

Critical Region :

Reject H_0 if $\chi^2_{cal} > \chi^2_{tab}$ where $\chi^2_{tab} = \chi^2_{\alpha, (r-1)(c-1)} = \chi^2_{0.05, (2-1)(3-1)} = \chi^2_{0.05, 2} = 5.991$

Table A.5 (continued) Critical Values of the Chi-Squared Distribution

ν	α						
	0.30	0.25	0.20	0.10	0.05	0.025	0.02
1	1.074	1.323	1.642	2.706	3.841	5.024	5.412
2	2.408	2.773	3.219	4.605	5.991	7.378	7.824
3	3.665	4.108	4.642	6.251	7.815	9.348	9.837

Calculation:

Gender	Non smoker	Moderate Smokers	Heavy Smokers	Total
Hypertension	$e_1 = \frac{o_1 = 21 \cdot 87 \cdot 69}{180} = 33.35$	$e_3 = \frac{o_3 = 36 \cdot 87 \cdot 62}{180} = 29.96$	$e_5 = \frac{o_5 = 30 \cdot 87 \cdot 49}{180} = 23.6$	87
Non Hypertension	$e_2 = \frac{o_2 = 48 \cdot 93 \cdot 69}{180} = 35.65$	$e_4 = \frac{o_4 = 26 \cdot 93 \cdot 62}{180} = 32.03$	$e_6 = \frac{o_6 = 19 \cdot 93 \cdot 49}{180} = 25.31$	93
Total	69	62	49	180

observed	Expected	$(o_i - e_i)$	$(o_i - e_i)^2$	$\frac{(o_i - e_i)^2}{e_i}$
21	33.35	-12.35	152.5225	4.57
48	35.65	12.35	152.5225	4.278
36	29.96	6.04	36.4816	1.2176
26	32.03	-6.03	36.3609	1.1352
30	23.68	6.32	39.9424	1.686
19	25.31	-6.31	39.8161	1.573

$$\chi^2_{cal} = 14.4598$$

14.4598

Conclusion :

$$\chi^2_{cal} > \chi^2_{tab} = 14.4598 > 5.991$$

Reject H_0 Hence we conclude that the hypertension & smoking habit are dependent to each other.

Q-Two hundred randomly selected adults were asked whether Tv shows as a whole are primarily entertaining, educational or a waste of time (only one answer could be chosen. The respondents were categorized by gender. Their responses are given in the following table. is this evidence convincing that there is a relationship between gender & opinion in the population interest?

Gender	Entertaining	Education	Waste of time	Total
Female	52	28	30	110
Male	28	12	50	90
Total	80	40	80	200

H_0 : The opinion of adults is independent of gender.

H_1 : The opinion of adults is dependent of gender.

Level of Significance: $\alpha = 0.05$

Test Statistics: $\chi^2 = \sum \left(\frac{(o_i - e_i)^2}{e_i} \right)$ $e_i = \frac{(\text{row total}) * (\text{column total})}{\text{Grand total}}$

Critical Region :

Reject H_0 if $\chi^2_{cal} > \chi^2_{tab}$ where $\chi^2_{tab} = \chi^2_{\alpha, (r-1)(c-1)} = \chi^2_{0.05, (2-1)(3-1)} = \chi^2_{0.05, 2} = 5.99$

Critical Region :

Reject H_0 if $\chi^2_{cal} > \chi^2_{tab}$ where $\chi^2_{tab} = \chi^2_{\alpha, (r-1)(c-1)} = \chi^2_{0.05, (2-1)(3-1)} = \chi^2_{0.05, 2} = 5.99$

Table A.5 (continued) Critical Values of the Chi-Squared Distribution

ν	α						
	0.30	0.25	0.20	0.10	0.05	0.025	0.02
1	1.074	1.323	1.642	2.706	3.841	5.024	5.412
2	2.408	2.773	3.219	4.605	5.991	7.378	7.824
3	3.665	4.108	4.642	6.251	7.815	9.348	9.837

Calculation:

Gender	Entertaining	Education	Waste of time	Total
Female	$e_1 = \frac{o_1 = 52 \times 80}{200} = 44$	$e_3 = \frac{o_3 = 28 \times 40}{200} = 22$	$e_5 = \frac{o_5 = 30 \times 80}{200} = 44$	110
Male	$e_2 = \frac{o_2 = 28 \times 80}{200} = 36$	$e_4 = \frac{o_4 = 12 \times 40}{200} = 18$	$e_6 = \frac{o_6 = 50 \times 80}{200} = 36$	90
Total	80	40	80	200

observed	Expected	$(o_i - e_i)$	$(o_i - e_i)^2$	$\frac{(o_i - e_i)^2}{e_i}$
52	44	8	64	1.454
28	22	6	36	1.636
30	44	14	196	4.455
28	36	-8	64	1.777
12	18	-6	36	2.000
50	36	14	196	5.444

$\chi^2_{cal} = 16.766$

16.766

Conclusion :

$\chi^2_{cal} > \chi^2_{tab} = 16.766 > 5.99$

Reject Ho Hence we conclude that the opinion of adults is not independent of gender

Q-In an industry 200 workers, employed for a specific job, were classified according to their performance & training received/ not received to test independence of a specific training & performance. The data is summarized as follows. Use chi square test of independence at 5% level of significance & write your conclusion.

	Good	Not Good	Total
Trained	100	50	150
Un trained	20	30	50
total	120	80	200

Solution:

H_0 :There is no association exist b/w performance & training workers

H_1 : There is an association exist b/w performance & training workers.

Level of Significance: $\alpha = 0.05$

Test Statistics: $\chi^2 = \sum \left(\frac{(o_i - e_i)^2}{e_i} \right)$ $e_i = \frac{(\text{row total}) * (\text{column total})}{\text{Grand total}}$

Critical Region :

Reject H_0 if $\chi^2_{cal} > \chi^2_{tab}$ where $\chi^2_{tab} = \chi^2_{\alpha, (r-1)(c-1)} = \chi^2_{0.05, (2-1)(2-1)} = \chi^2_{0.05,1} = 3.841$

Table A.5 (continued) Critical Values of the Chi-Squared Distribution

ν	α						
	0.30	0.25	0.20	0.10	0.05	0.025	0.02
1	1.074	1.323	1.642	2.706	3.841	5.024	5.412
2	2.408	2.773	3.219	4.605	5.991	7.378	7.824

Calculation :

	Good	Not Good	Total
Trained	$e_1 = \frac{o_1 = 100}{150 * 50} = 90$	$e_2 = \frac{o_2 = 50}{150 * 80} = 60$	150
Un trained	$e_3 = \frac{o_3 = 20}{50 * 120} = 30$	$e_4 = \frac{o_4 = 30}{50 * 80} = 20$	50
total	120	80	200

o_i	e_i	$(o_i - e_i)^2$	$\frac{(o_i - e_i)^2}{e_i}$
100	90	100	1.11
50	60	100	1.66
20	30	100	3.33
30	20	100	5

11.11

$$\chi^2_{cal} = 11.11$$

Conclusion :

$$\chi^2_{cal} > \chi^2_{tab} = 11.11 > 3.841$$

Reject H_0 Hence we conclude that there is an association exist b/w performance & training workers.

TESTING A POPULATION VARIANCE

Let us first consider the problem of testing the null hypothesis , H_0 That the population variance σ^2 equal a specified value σ_0^2 against one of usual alternative.

1- $\sigma^2 < \sigma_0^2$

2- $\sigma^2 > \sigma_0^2$

3- $\sigma^2 \neq \sigma_0^2$

The appropriate statistic Chi-Square is used:

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

Where ,

n is the random sample drawn from normal population

S^2 is the sample variance

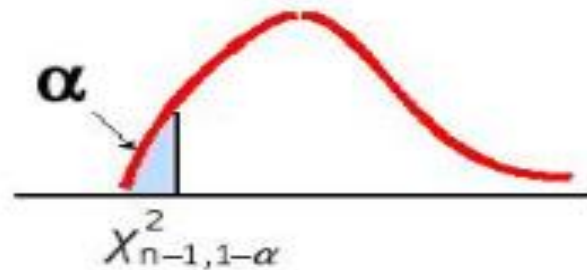
Decision Rules: Variance

Population variance

Lower-tail test:

$$H_0: \sigma^2 \geq \sigma_0^2$$

$$H_1: \sigma^2 < \sigma_0^2$$

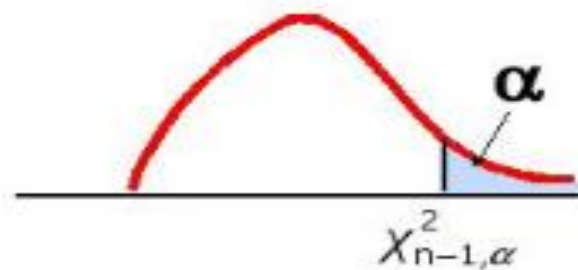


Reject H_0 if
 $\chi_{n-1}^2 < \chi_{n-1, 1-\alpha}^2$

Upper-tail test:

$$H_0: \sigma^2 \leq \sigma_0^2$$

$$H_1: \sigma^2 > \sigma_0^2$$

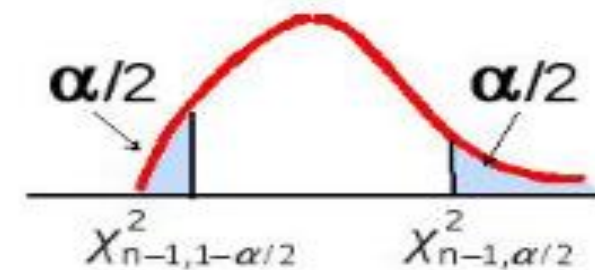


Reject H_0 if
 $\chi_{n-1}^2 > \chi_{n-1, \alpha}^2$

Two-tail test:

$$H_0: \sigma^2 = \sigma_0^2$$

$$H_1: \sigma^2 \neq \sigma_0^2$$



Reject H_0 if
 $\chi_{n-1}^2 > \chi_{n-1, \alpha/2}^2$
or
 $\chi_{n-1}^2 < \chi_{n-1, 1-\alpha/2}^2$

H_0	H_1	Test statistic	Critical Region
$\sigma^2 = \sigma_0^2$	$\sigma^2 \neq \sigma_0^2$	$\chi^2 = \frac{S^2(n-1)}{\sigma^2}$	$\chi^2 < \chi_{\alpha/2, \nu}^2$ and $\chi^2 > \chi_{1-\alpha/2, \nu}^2$
	$\sigma^2 > \sigma_0^2$		$\chi^2 > \chi_{1-\alpha, \nu}^2$
	$\sigma^2 < \sigma_0^2$		$\chi^2 < \chi_{\alpha, \nu}^2$

where

$\chi_{1-\alpha, \nu}^2$ = upper-tail value of the r.v. χ^2 for $1 - \alpha$ and ν degrees of freedom

$\chi_{\alpha, \nu}^2$ = lower-tail value of the r.v. χ^2 for α and ν degrees of freedom

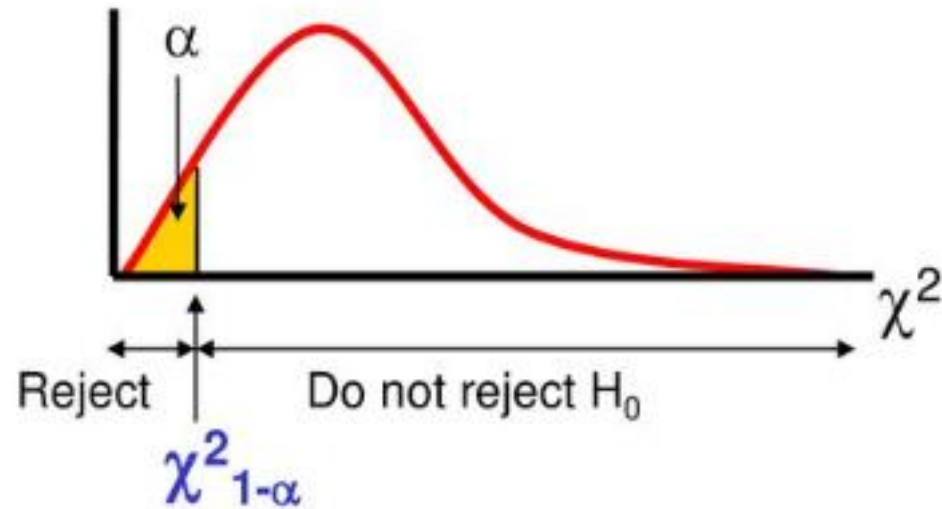
ν = $n - 1$ = degrees of freedom

n = sample size

Lower tail test:

$$H_0: \sigma^2 \geq \sigma_0^2$$

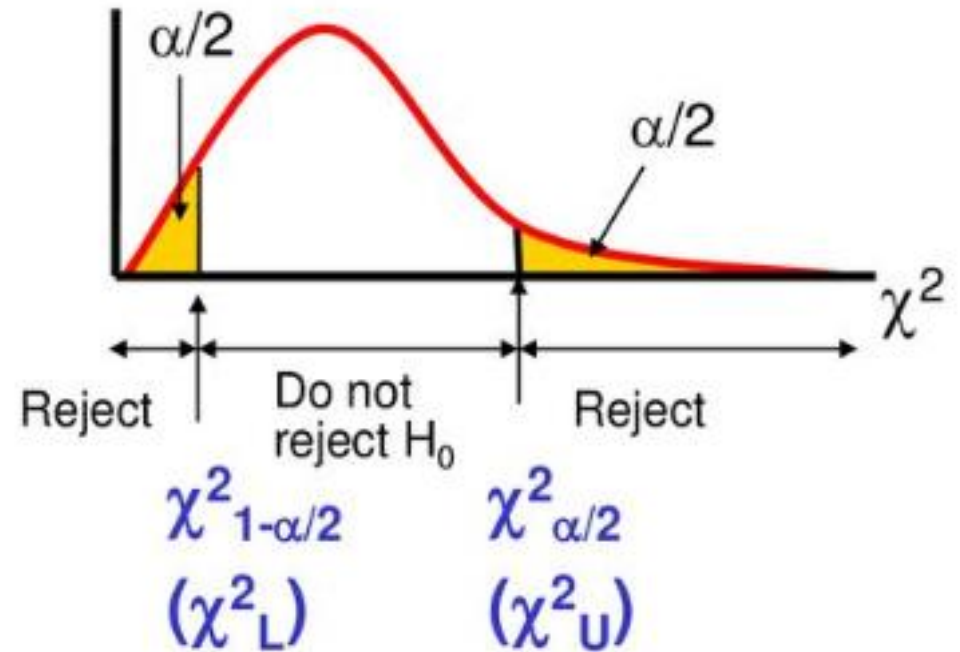
$$H_A: \sigma^2 < \sigma_0^2$$



Two tail test:

$$H_0: \sigma^2 = \sigma_0^2$$

$$H_A: \sigma^2 \neq \sigma_0^2$$



10.68 Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a standard deviation of 6 minutes. Test, the hypothesis that $\sigma = 6$ against the alternative that, $\sigma < 6$ if a random sample of 20 high school seniors has a standard deviations $s = 4.51$. Use a 0.05 level of significance?

Solution:

Step-1

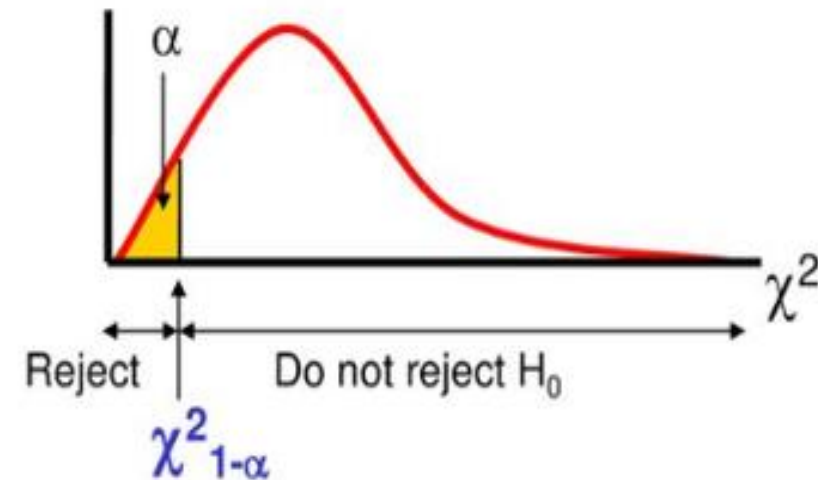
$$H_0 : \sigma = 6 \quad H_1 : \sigma < 6$$

Step – 2 (Level of Significance)

$$\alpha = 0.05$$

Step – 3 (Test Statistic)

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$



Step – 4 (Critical Region)

Reject H_0 if $\chi^2_{cal} < \chi^2_{tab}$

$$\chi^2_{cal} < \chi^2_{(1-\alpha, n-1)}$$

$$\chi^2_{cal} < \chi^2_{(1-0.05, 20-1)}$$

$$\chi^2_{cal} < \chi^2_{(0.95, 19)}$$

if $\chi^2_{cal} < 10.117$

Step – 5 Calculation

$S=4.51$, $n=20$, $\sigma = 6$

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{(20-1)(4.51)^2}{(6)^2} = 10.73$$

Step – 6 (Decision)

$10.73 < 10.117$ **Accept H_0 & conclude that $\sigma = 6$**

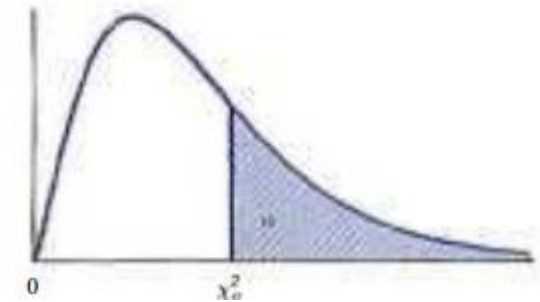


Table A.5 Critical Values of the Chi-Squared Distribution

ν	α				
	0.995	0.99	0.98	0.975	0.95
18	6.265	7.015	7.906	8.231	9.390
19	6.844	7.633	8.567	8.907	10.117
20	7.434	8.260	9.237	9.591	10.851

10.69 Allotoxins produced by mold on peanut crops in Virginia must be monitored. A sample of 61 batches of peanuts reveals levels of 24.17 ppm, on average, with a variance of 4.25 ppm. Test the hypothesis that $\sigma^2 = 4.2$ ppm with the alternative that $\sigma^2 \neq 4.2$ ppm.

Solution:

Step-1

$$H_0 : \sigma^2 = 4.2 \quad H_1 : \sigma^2 \neq 4.2$$

Step – 2 (Level of Significance)

$$\alpha = 0.05$$

Step – 3 (Test Statistic)

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

Step – 4 (Critical Region)

Reject H_0 if $\chi^2_{cal} < \chi^2_{tab}$ or $\chi^2_{cal} > \chi^2_{tab}$

$$\chi^2_{cal} < \chi^2_{(1-\frac{\alpha}{2}, n-1)} \qquad \chi^2_{cal} > \chi^2_{(\frac{\alpha}{2}, n-1)}$$

$$\chi^2_{cal} < \chi^2_{(1-\frac{0.05}{2}, 61-1)} = \chi^2_{(0.975, 60)} = 40.482 \qquad \text{or} \qquad \chi^2_{cal} > \chi^2_{(\frac{0.05}{2}, 61-1)} = \chi^2_{(0.025, 60)} = 83.298$$

Reject H_0 if $\chi^2_{cal} < 40.482$ or $\chi^2_{cal} > 83.298$

ν	0.995	0.99	0.98	0.975
40	20.707	22.164	23.838	24.433
50	27.991	29.707	31.664	32.357
60	35.534	37.485	39.699	40.482

ν	α		
	0.10	0.05	0.025
40	51.805	55.758	59.342
50	63.167	67.505	71.420
60	74.397	79.082	83.298

Step – 5 Calculation

$$S^2=4.25, n = 61, \sigma^2 = 4.2$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{(61-1)(4.25)}{4.2} = 63.75$$

Step – 6 (Decision)

$$63.75 < 40.482 \text{ or } 63.75 > 83.298 \text{ Accept } H_0 \text{ \& conclude that } \sigma^2 = 4.2$$

10.71 A soft-drink dispensing machine is said to be out of control if the variance of the contents exceeds 1.15 deciliters. If a random sample of 25 drinks from this machine has a variance of 2.03 deciliters, does this indicate at the 0.05 level of significance that the machine is out of control? Assume that the contents are approximately normally distributed.

Solution:

Step-1

$$H_0 : \sigma^2 \leq 1.15 \quad H_1 : \sigma^2 > 1.15$$

Step – 2 (Level of Significance)

$$\alpha = 0.05$$

Step – 3 (Test Statistic)

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

Step – 4 (Critical Region)

Reject H_0 if $\chi^2_{cal} > \chi^2_{tab}$ $\chi^2_{cal} > \chi^2_{(\alpha, n-1)}$

$$\chi^2_{cal} > \chi^2_{(0.05, 25-1)} \qquad \chi^2_{cal} > \chi^2_{(0.05, 24)}$$

if $\chi^2_{cal} > 36.415$

Step – 5 Calculation

$$S^2 = 2.03, n = 25, \sigma^2 = 1.15$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{(25-1)(2.03)}{1.15} = 42.36$$

Table A.5 (continued) Critical Values of the Chi-Squared Distribution

ν	0.30	0.25	0.20	0.10	0.05
23	26.018	27.141	28.429	32.007	35.172
24	27.096	28.241	29.553	33.196	36.415
25	28.172	29.339	30.675	34.382	37.652

Step – 6 (Decision)

$42.36 > 36.415$ Reject H_0 & conclude that $\sigma^2 > 1.15$

10.67 The volume of containers of a particular lubricant is known to be normally distributed with a variance of 0.03 liter, Test the hypothesis that a $\sigma^2 = 0.03$ against the alternative that a $\sigma^2 \neq 0.03$ for the random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal.

Solution:

Step-1

$$H_0 : \sigma^2 = 0.03 \quad H_1 : \sigma^2 \neq 0.03$$

Step – 2 (Level of Significance)

$$\alpha = 0.01$$

Step – 3 (Test Statistic)

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

Step – 4 (Critical Region)

Reject H_0 if $\chi^2_{cal} < \chi^2_{tab}$ or $\chi^2_{cal} > \chi^2_{tab}$

$\chi^2_{cal} < \chi^2_{(1-\frac{\alpha}{2}, n-1)}$ or $\chi^2_{cal} > \chi^2_{(\frac{\alpha}{2}, n-1)}$

$\chi^2_{cal} < \chi^2_{(1-\frac{0.01}{2}, 10-1)} = \chi^2_{(0.995, 9)} = 1.735$ or $\chi^2_{cal} > \chi^2_{(\frac{0.01}{2}, 10-1)} = \chi^2_{(0.005, 9)} = 23.589$

Reject H_0 if $\chi^2_{cal} < 1.735$ or $\chi^2_{cal} > 23.589$

ν	0.01	0.005	0.001
7	18.475	20.278	24.321
8	20.090	21.955	26.124
9	21.666	23.589	27.877

ν	0.995	0.99
1	0.004393	0.003157
2	0.0100	0.0201
3	0.0717	0.115
4	0.207	0.297
5	0.412	0.554
6	0.676	0.872
7	0.989	1.239
8	1.344	1.647
9	1.735	2.088

Step – 5 Calculation

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}, \quad \bar{x} = \frac{\sum x_i}{n} = 10.06$$

$$S = \sqrt{\frac{\sum (x_i - 10.06)^2}{10-1}} = 0.245$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{(10-1)(0.245)^2}{0.03} = 18.0075$$

Step – 6 (Decision)

$$18.0075 < 1.735 \quad \text{or} \quad 18.0075 > 23.589$$

Accept H_0 & conclude that $\sigma^2 = 0.03$

Example:

- A commercial freezer must hold the selected temperature with little variation. Specifications call for a standard deviation of no more than 4 degrees (or variance of 16 degrees²).
- A sample of 16 freezers is tested and yields a sample variance of $s^2 = 24$. Test to see whether the standard deviation specification is exceeded. Use $\alpha = 0.05$



- Use the chi-square table to find the critical value:

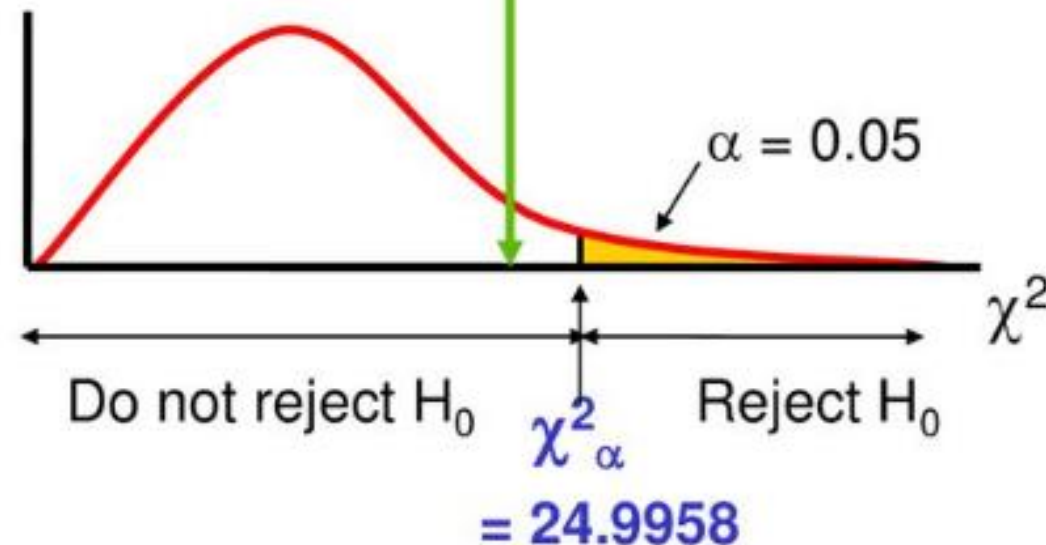
$$\chi^2_{\alpha} = 24.9958 \quad (\alpha = 0.05 \text{ and } 16 - 1 = 15 \text{ d.f.})$$

The test statistic is:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(16-1)24}{16} = 22.5$$

**Since $22.5 < 24.9958$,
do not reject H_0**

There is not significant evidence at the $\alpha = 0.05$ level that the standard deviation specification is exceeded



Thank You!